CellOracle HIF1A knockout: what generates what

Repository guide for HIF-1alpha integrates metabolic and immunoregulatory programs in RORgt+ regulatory T cells during intestinal inflammation (Cipelli et al.).

START	CORE ANALYSIS	DOWNSTREAM OUTPUTS
Seurat export counts + genes + cells + metadata + UMAP	run_hif1a_knockout.py Oracle + cluster GRNs + HIF1A = 0 simulation	ForceAtlas figures Perturbation score + Markov Propagation impact

Execution map

Order	Executable	Main inputs	What it calculates	What it writes	Use
1	**run_hif1a_ knockout.py**	Seurat export; promoter base-GRN	AnnData import; PCA/KNN; cluster-specific Links; GRN fitting; HIF1A=0; transition probabilities; UMAP shifts	Oracle objects; Links; gene/cell/cluster delta tables; UMAP, quiver, grid flow and propagation figures	Required first step. Computationally heavy.
2	**generate_FA_ paperstyle_HIF1A_KO.py**	Simulated Oracle; cell_delta_metrics.csv	ForceAtlas embedding; L2 magnitude; cell shifts; smoothed grid flow; optional local pseudotime-alignment score	FA PNG/PDF panels; embedding/vector CSVs; propagation L2 table; Oracle with FA coordinates	Use for ForceAtlas panels after step 1.
3	**generate_PS_ markov_HIF1A_KO.py**	Simulated Oracle with UMAP, Pseudotime and group metadata	Pseudotime gradient; CellOracle perturbation score; Markov trajectories; endpoint density and transitions	PS/Markov PNG/PDF panels; grid and cluster summaries; trajectory NPZ; gradient object; logs	Use for trajectory-effect interpretation.
4	**regenerate_ propagation_HIF1A_KO.py**	Existing simulated Oracle	Only propagation steps 0-5 using CellOracle's default L1 delta-X length	Final propagation PNG/PDF and tidy CSV by cluster and propagation step	Fast figure-only rerun; does not repeat GRN/PCA.

Rule of thumb: scripts 2-4 all start from the Oracle produced by script 1. They are downstream branches, not sequential prerequisites for one another.

Files generated by each script

The table below lists the files most useful for auditing, figure regeneration and data sharing.

Script	Objects	Tables	Figures	Logs / provenance
run_hif1a_ knockout.py	input_for_celloracle.h5ad oracle_imported...oracle oracle_preprocessed...oracle links_filtered...links oracle_fitted_grn...oracle oracle_HIF1A_KO...oracle	base_grn_used.csv links_raw.csv links_filtered.csv network_scores.csv gene_delta_summary.csv cell_delta_metrics.csv cluster_delta_summary.csv	umap_cluster_and_target.png hif1a_ko_quiver_cells.png hif1a_ko_grid_arrows.png n_propagation_impact.png/.pdf	run_info.json
generate_FA_ paperstyle_HIF1A_KO.py	oracle_HIF1A_KO_FA...oracle	fa_embedding_cells.csv cell_shift_vectors_FA.csv grid_shift_vectors_FA.csv propagation_magnitude_by_cluster.csv perturbation_score_FA.csv (when pseudotime exists)	FA clusters cell arrows + sampled arrows grid flow delta-L2 map/distributions L2 propagation FA perturbation-score map	summary.json run_parameters.json ForceAtlas fallback note, if used
generate_PS_ markov_HIF1A_KO.py	HIF1A_KO_markov_trajectories.npz HIF1A_KO_pseudotime_gradient...gradient	perturbation_score_grid.csv perturbation_score_by_cluster.csv markov_start_end_cells.csv markov_density_by_cell.csv transition counts/proportions endpoint cluster by group	HIF1A_KO_perturbation_score_PS HIF1A_KO_markov_simulation_density combined PS + Markov PS + Markov by group (PNG and PDF)	summary.json run_parameters.json methodology.txt
regenerate_ propagation_HIF1A_KO.py	Reads Oracle; writes no new Oracle	HIF1A_KO_propagation_mean_delta_x_length_by_cluster.csv	HIF1A_KO_propagation_magnitude_by_cluster.png/.pdf	Prints absolute output paths

What to share with the paper

Share	Recommended content	Reason
Code repository	Four executable scripts, functions/, environment.yml, requirements file and this README/PDF	Allows full reproduction from the exported single-cell input
Figure source data	CSV tables underlying propagation, perturbation score, Markov summaries and embedding vectors	Lets readers reproduce panels without rerunning the expensive GRN fit
Data archive	Final Oracle and Links objects, if repository size permits; otherwise deposit externally with checksums	Preserves fitted networks and simulated state
Do not treat as counts	Vector fields, perturbation score and Markov endpoint density	These represent predicted state dynamics, not post-perturbation cell abundance

Scientific interpretation and reproducibility notes

Quantity	Definition in this repository	Interpretation	Do not confuse with
Mean delta X length	Mean per-cluster L1 norm of simulated delta-X, evaluated across propagation steps	Magnitude of the propagated transcriptional response	Euclidean/L2 distance
delta_l2 / mean_l2_norm	Euclidean (L2) norm of the simulated expression-shift vector	Predicted expression-space displacement magnitude	CellOracle default L1 propagation-impact plot
Perturbation score	Local inner product between HIF1A-KO simulation flow and pseudotime-gradient flow	Positive: alignment; negative: opposition to observed progression	Probability, effect size or cell count
Markov endpoint density	Relative KDE of simulated trajectory endpoints, scaled 0-1 after percentile clipping	Relative predicted accumulation tendency in embedding space	Absolute post-perturbation abundance

Reusable modules

Reusable module	Responsibility	Used by
functions/io_utils.py	Cache paths and output-directory creation	All pipelines
functions/celloracle_utils.py	Oracle loading/validation, Seurat-export import, GRN loading and table export	Core, FA, PS/Markov and propagation
functions/embedding_utils.py	ForceAtlas recovery/calculation, embedding tables and shift/grid vectors	FA pipeline
functions/perturbation_utils.py	Propagation summaries, pseudotime score, CellOracle PS and Markov calculations	Core downstream analyses
functions/plotting_utils.py	Publication-ready plots, fixed cluster palette and PNG/PDF export	All figure-producing scripts

Minimal rerun commands

```
1) conda run -n celloracle-hif1a python code/Figure_6/figure_6D_E_F/celloracle/run_hif1a_knockout.py ...
2) conda run -n celloracle-hif1a python code/Figure_6/figure_6D_E_F/celloracle/generate_FA_paperstyle_HIF1A_KO.py
3) conda run -n celloracle-hif1a python code/Figure_6/figure_6D_E_F/celloracle/generate_PS_markov_HIF1A_KO.py
4) conda run -n celloracle-hif1a python code/Figure_6/figure_6D_E_F/celloracle/regenerate_propagation_HIF1A_KO.py
```

Scope statement

The in silico knockout sets HIF1A expression to zero and propagates the predicted shift through cluster-specific regulatory models. Embedding vectors and perturbation scores describe predicted changes in cell-state dynamics. They do not directly estimate proliferation, survival or post-perturbation cell numbers.